

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Current Electricity - Core Application	52 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Current Electricity - Core Application (Very High Yield — 52 PYQs)

Core Principle: Key problem pattern in Current Electricity.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Refer to the figure given below. The values of I_1 , I_2 and I_3 are ____" [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "Two resistors of $200\ \Omega$ and $400\ \Omega$ are connected in series with a battery of 100 V . A bulb rated at 200 V , 100 W is connected ac..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "In the potentiometer, when the cell in the secondary circuit is shunted with $4\ \Omega$ resistance, the balance is obtained at the length 120 cm ..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Refer to the figure given below. The values of I_1 , I_2 and I_3 are ____ .

- (A) $I_1=2.5A, I_2=1.875A, I_3=1.875A$ **[Correct]**
- (B) $I_1=1.875A, I_2=2.5A, I_3=1.875A$
- (C) $I_1=1.875A, I_2=1.875A, I_3=2.5A$
- (D) $I_1=2.5A, I_2=2.5A, I_3=1.875A$

Correct Answer: A

Solution Analysis: Using Kirchhoff's current law, the net accumulation of current at a junction is zero. So, at junction A, $I_1 - I_3 - I_{AD} = 0 \Rightarrow I_{AD} = I_1 - I_3$ And at junction D, $I_{AD} + I_2 - I_{DB} = 0 \Rightarrow I_1 - I_3 + I_2 - I_{DB} = 0 \Rightarrow I_{DB} = I_1 - I_3 + I_2$ Applying Kirchhoff's voltage law in closed loop I, $-4(I_1 - I_3) - 2(I_1 + I_2 - I_3) - I_1 + 10 = 0$
 $-7I_1 - 2I_2 + 6I_3 = -10 \dots (i)$ In loop (II) $I_1 + 10 - 4I_3 - 2(I_3 - I_2) = 0$ $2I_2 - 6I_3 - I_1 = -10 \dots (ii)$ In loop III,
 $-5 - 2(I_3 - I_2) + 2(I_1 + I_2 - I_3) = 0$ $2I_1 + 4I_2 - 4I_3 = 5 \dots (iii)$ From e